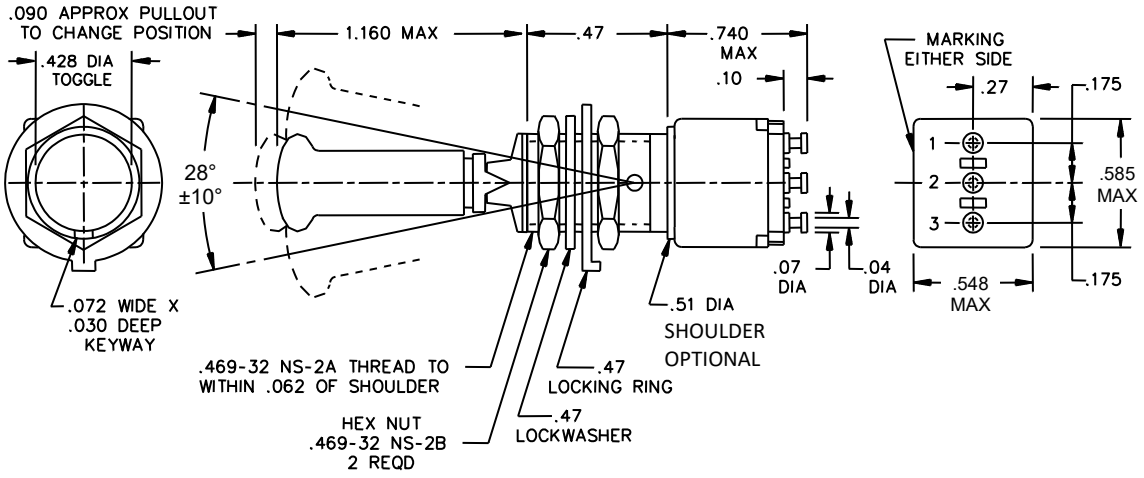
INCH-POUND
MS27720J
21 October 2021
SUPERSEDING
MS27720H
w/Amendment 1
25 May 2016

DETAIL SPECIFICATION SHEET

SWITCH, TOGGLE, MINIATURE, SINGLE POLE TOGGLE SEAL, LEVER LOCK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the latest issue of MIL-DTL-83731.



Inches	mm	Inches	mm	Inches	mm
.005	0.13	.072	1.83	.47	11.94
.020	0.51	.090	2.29	.51	12.95
.030	0.76	.10	2.54	.535	13.59
.04	1.02	.175	4.45	.575	14.61
.062	1.57	.27	6.86	.740	18.80
.07	1.78	.428	10.87	1.160	29.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.020 (0.51 mm) on two place decimals and ±.005 (0.13 mm) on three place decimals.
4. Configuration of switch case is optional provided maximum dimensions specified are not exceeded.
5. For hardware detail specifications, see supplement of MIL-DTL-83731.

FIGURE 1. Dimensions and configurations.

AMSC N/A
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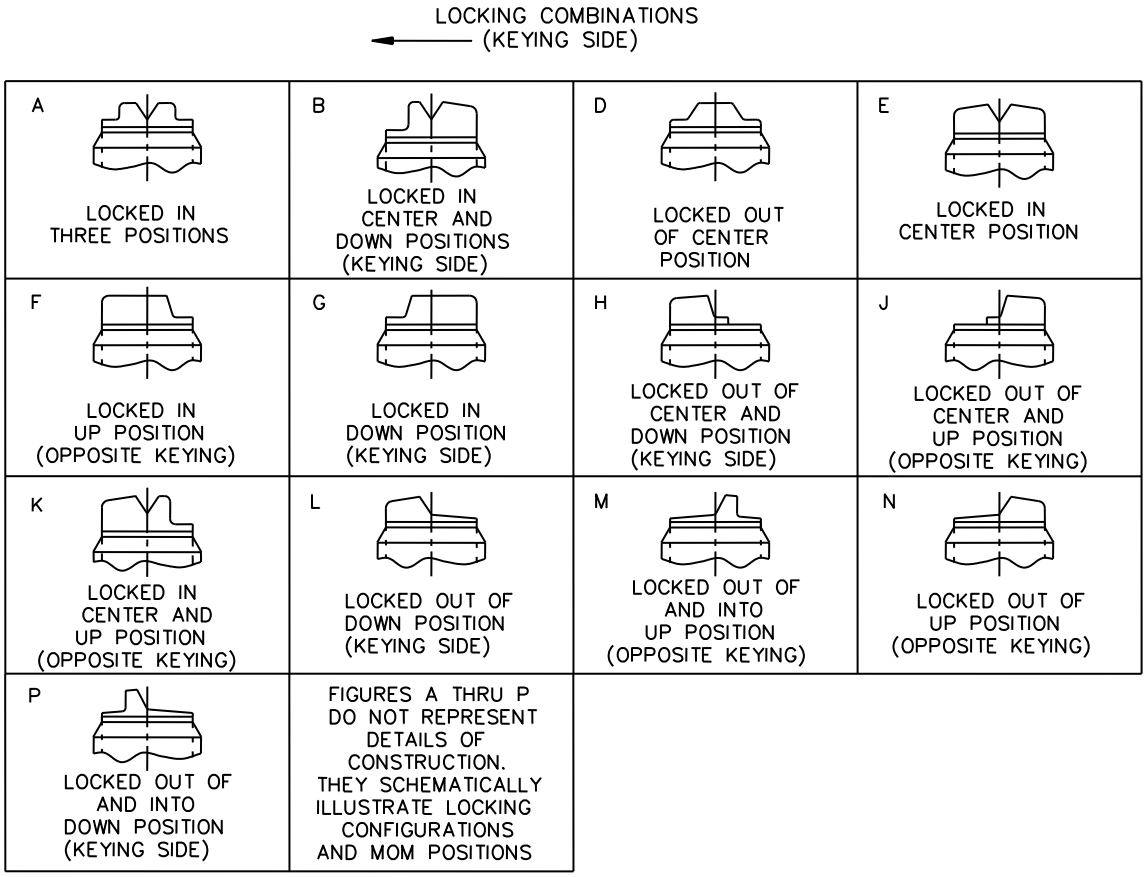


FIGURE 2. Detail requirements

Canceled MS part number 1/	Substitute part number 1/	Available locking combinations	Circuits made with toggle lever in		
			Keying side 1-2	Center position	Opposite keying side 2-3
MS27720-21	MS27720-21-1	All	ON	OFF	ON
-22	-22-1	D, F, G	OFF	NONE	ON
-23	-23-1	D, F, G	ON	NONE	ON
-26	-26-1	F	MOM-ON	NONE	ON
-27	-27-1	E, L, N	MOM-ON	OFF	MOM-ON
-31	-31-1	E, F, K, L, M, N	MOM-ON	OFF	ON

1/ The canceled part numbers are no longer acquirable; the substitute part numbers have a lower dielectric withstanding voltage at reduced barometric pressure (see test voltage requirement below.)

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REQUIREMENTS:

All switches on this standard are designed so that the movement of the switch mechanism is opposite to that of the toggle lever.

Test voltage for dielectric withstanding voltage at reduced barometric pressure, shall be 400 V rms minimum.

Electrical endurance shall be performed for 30,000 cycles.

Mechanical endurance shall be performed 50,000 cycles at -65°C and then 50,000 cycles at 85°C.

Intermediate current shall be performed for 30,000 cycles.

Terminals shall be molded into the switch case and shall accommodate No. 18 AWG wire.

Locking arrangement: Positive locking shall be accomplished and shall prevent motion of the toggle lever until the locking mechanism is manually released.

The force required to release the locking mechanism shall be 3 to 5 pounds.

The locking means at the top of the toggle bushing shall be capable of withstanding a torque of 20 pound inches applied in both directions immediately following the humidity test.

Maximum weight shall be .066 pound.

Electrical ratings for 28 V dc and 115 V ac at 400 Hz. Lamp load = 1 ampere, resistive load = 5 amperes, inductive load = 2 amperes.

Part or Identifying Number (PIN) example: MS27720-21-1A (locking combination 'A').

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Referenced documents
MIL-DTL-83731

The margins of this specification are marked with vertical lines to indicate where modifications from this revision were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodian:
Army - CR
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5930-2021-044)

Review activity:
Army - AV

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